

Total no. of Pages...01.

Roll no.

THIRD SEMESTER B. Tech.(ECAM)
MID-SEMESTER EXAMINATION, SEPTEMBER 2024

Course Code- **EAEP C302**

Course Title- **Signals and Systems**

Time- 1.5 Hours

Max. Marks- 20

Note: - Attempt all questions. Missing data/ information if any, maybe suitably assumed & mentioned in the answer.

Q. No.	Questions	Marks	CO
1a	Define different types of signals and systems.	2	CO 1
1b	How an analog signal is converted to a digital signal?	1	
1c	Draw the function $\pi(3t+2)$ when $\pi(t) = 2$; for $t \leq 1/2$, 0 ; otherwise	2	
2a	Define following signals-(a) Memoryless (b) time invariant (c)linear (d)stable	2	CO1
2b	Determine whether the following discrete -time signal is periodic determine fundamental period. $X[n]=\sin((6\pi/7)n + 1)$	2	
3a	Determine whether each of the following statements concerning LTI is true or false justify your answers (a)the inverse of a causal LTI system is always a causal. (b)If an LTI system is causal it is stable.	2	CO2
3b	Name some properties of an LTI systems with example.	2	CO2
4a	Express each of the following complex numbers in polar form and plot them in complex plane indicating magnitude and angle (i) $j(1+j)e^{j\pi/6}$ (ii) $(1-j\sqrt{3})^3$	1	CO1
4b	The following are the impulse responses of the continuous time LTI systems. Determine whether each system is causal or stable. Justify your answer $h(t)=e^{-4t}u(t-2)$ $h(t)=e^{2t}u(-1-t)$	2	CO2
5b	In $y[n]=x[n]x[n-2]$ (a)Is the system memoryless ? (b)determine output when input is $A\delta[n]$. (c)Is the system invertible?	2	CO2
5a	Let $x(t)=u(t-3)-u(t-5)$ and $h(t)=e^{-3t}u(t)$ compute $y(t)=x(t)*h(t)$	2	CO2